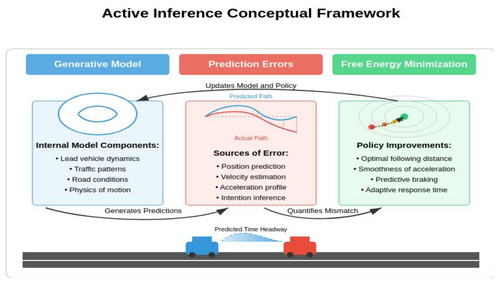
Active Inference for Car-Following: A Real-Time Approach to Safe Time Headway Maintenance in CARLA

Varadaraya Ganesh Shenoy, Theodore Morris, Vassilios Morellas

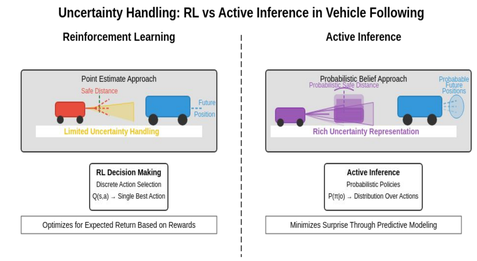
University of Minnesota, Minneapolis, MN

Introduction

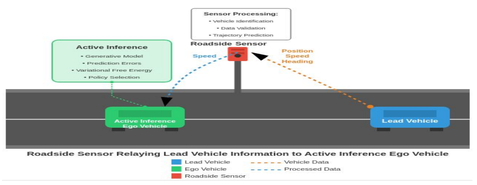
- **Active inference frameworks** are **pivotal for developing robust autonomous vehicle control systems**.



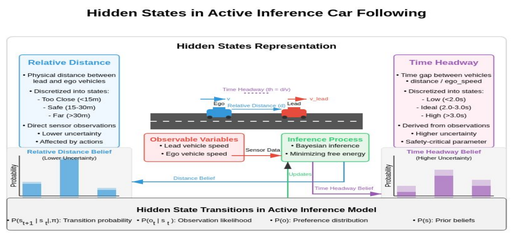
- **Generative models** allow vehicles to **predict and respond** proactively **unlike reinforcement learning approaches**, which learn through trial and error to maximize rewards



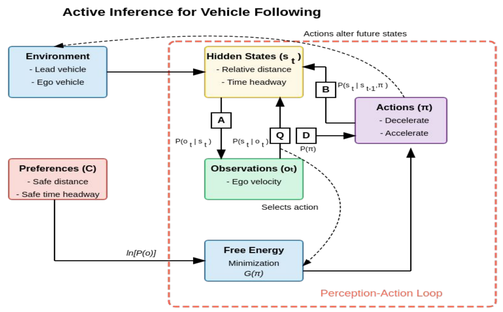
- A **roadside sensor-based** vehicle following system that utilizes **active inference principles**. The flow of information works as follows:
  - The lead vehicle (blue) transmits vehicle data to the roadside sensor.
  - The roadside sensor processes this raw data to perform **vehicle identification, data validation, and trajectory prediction**. This processed information is then relayed to the ego vehicle (green).



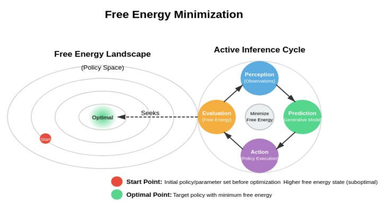
Active Inference Framework



- **Observation Model (A)**: Connects **hidden states** to **sensory data** by defining **$P(o_t|s_t)$**. This helps the system predict what its sensors should detect if its beliefs are correct.
- **Transition Model (B)**: Predicts how the **world will change through $P(s_t|s_{t-1},\pi)$** - how the current state and actions lead to future states. This allows planning by simulating outcomes of potential actions.
- **Preferences (C)**: Encoded as **prior beliefs** about **desired observations,** representing the system's goals.
- **Prior Beliefs (D):** Encodes **prior beliefs** about **hidden states** before receiving observations.

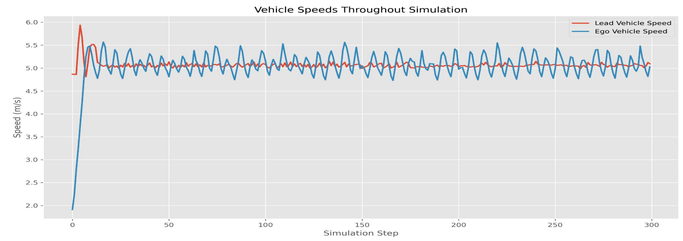


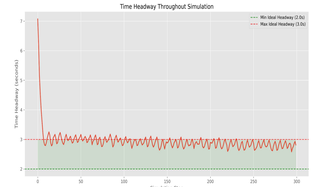
- The **active inference cycle** consists of four key components – **perception**, **prediction**, **evaluation** and **action** (executing policies) – that work together to continuously r**educe surprise** and converge on **optimal behavior**. Thus, allowing the system to move from an **initial high-energy** state toward a **minimum free energy** landscape.

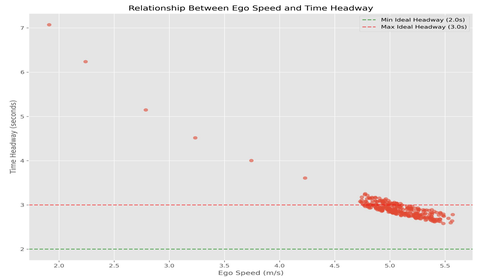


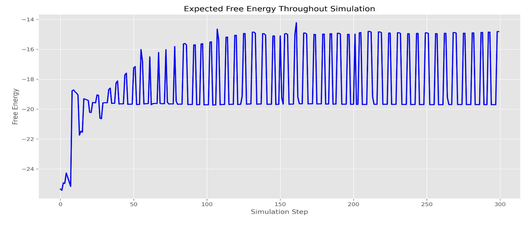
Simulation test on CARLA

Results for Vehicle Following with constant speed

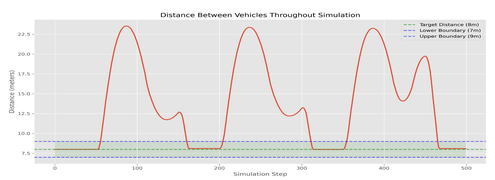








Results for Vehicle Following for constant speed behaviour





Results for Vehicle Following for stop and go behaviour

Conclusions and Future Work

- The **model successfully** maintains **time headway** even during **stop-and-go traffic scenarios**, with the ego vehicle adapting to the lead vehicle's changing behavior.
- The system demonstrates **experience-based learning,** as each **interaction** provides **refines** both **transition** and **observation models.**
- Future research will focus on **enabling continuous state representation** for time headway, lead vehicle speed, and ego vehicle speed,while **improving predictive accuracy** by representing uncertainty through continuous probability density functions, allowing for more **nuanced probabilistic reasoning** and **continous control values.**
- Extend the model to handle **vulnerable road user behaviour** by combining the **Active Inference framework with perception models** and integrate **contextual factors** (crosswalks, traffic signals) that influence the **behaviour of road users**.

Acknowledgements

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